

Liver cirrhosis



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Learning Objectives

Upon completion of this lecture , the student is able to:

Recognize the normal physiologic functions of the liver, including its role in bile formation and secretion, metabolism, vitamin storage, coagulation, and detoxification.

Discuss diagnosis of specific pathologic conditions that affect the liver, including cirrhosis, hepatitis,.

Recognize complications of liver decompensation, including, portal hypertension, esophageal varices

Recognize dental aspects of liver diseases

Functions of the Liver

Metabolism / detoxification

Metabolizes products of digestion

- Glucose regulation
- Vitamin storage
- Metabolizes drugs
- Breaks down bilirubin

storage and Filtration of Blood

- Acts as a blood reservoir
- Contains phagocytic Cells
- Part of the reticuloendothelial system.

Synthesis and Secretion

- Components of clotting factors
- Cholesterol, triglyceride synthesis
- Bile production
- Other proteins and hormones

Liver cirrhosis

Definition: **Necrosis** of liver cells followed by **fibrosis** and **nodule** formation.

Loss of normal liver **architecture** interferes with liver **blood flow and function**.

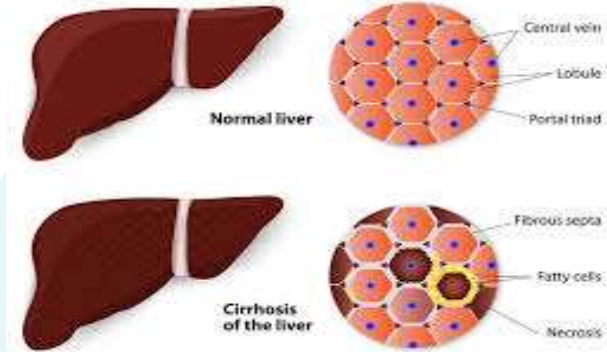
Stage 1 cirrhosis:

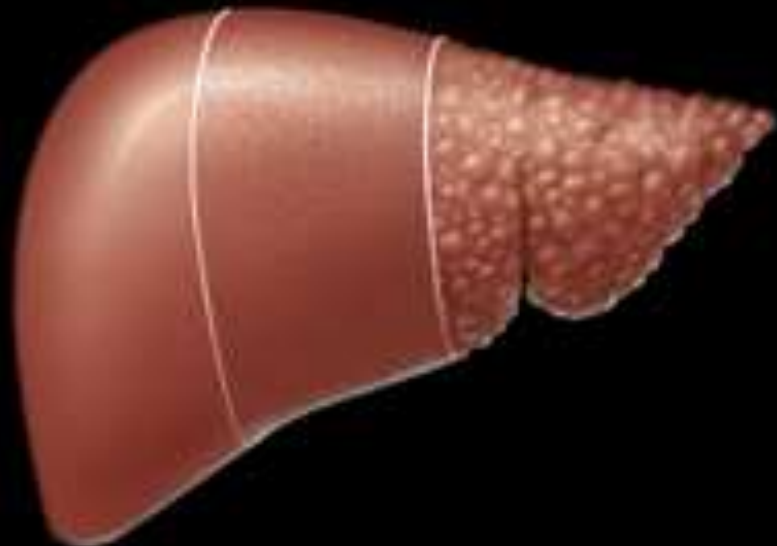
Compensated and asymptomatic

Can last for 10-20 year

Almost normal life expectancy

Stage 2 cirrhosis is the onset of first de-compensation

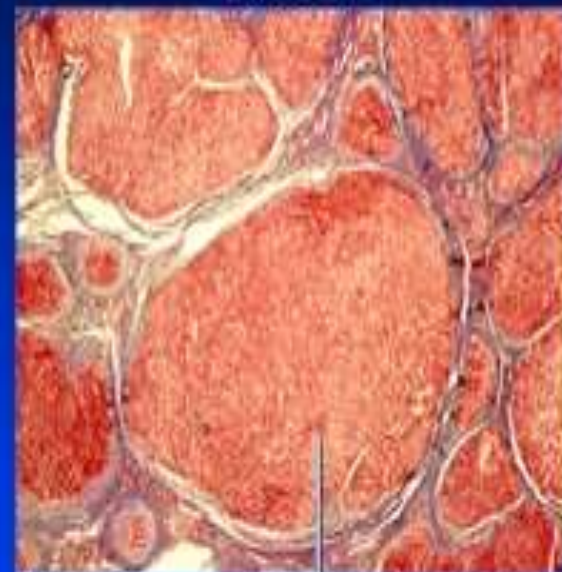




Normal



Cirrhosis



Nodules surrounded by
fibrous tissue

NORMAL LIVER VS LIVER CIRRHOSIS

Causes:

1

Alcoholism

2

Viral infection caused by Chronic Hepatitis B & C

3

Fat accumulation in the liver

4

Iron buildup in the body

5

Copper accumulation in the liver (Wilson's disease)

6

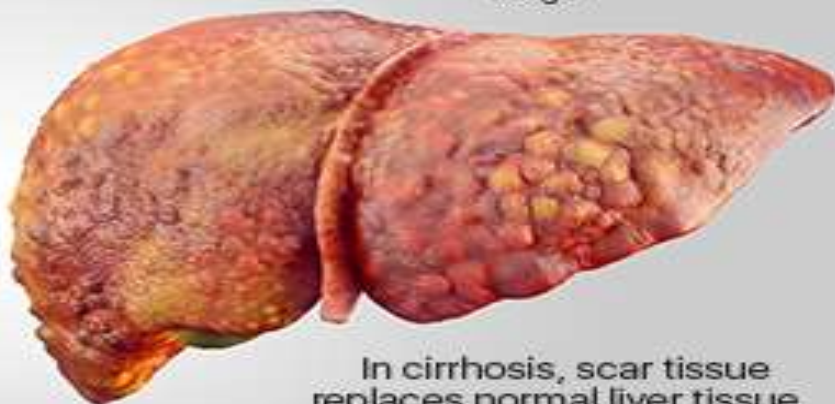
Toxic Hepatitis caused by severe reactions to prescribed drugs

7

Repeated bouts of heart failure with liver congestion

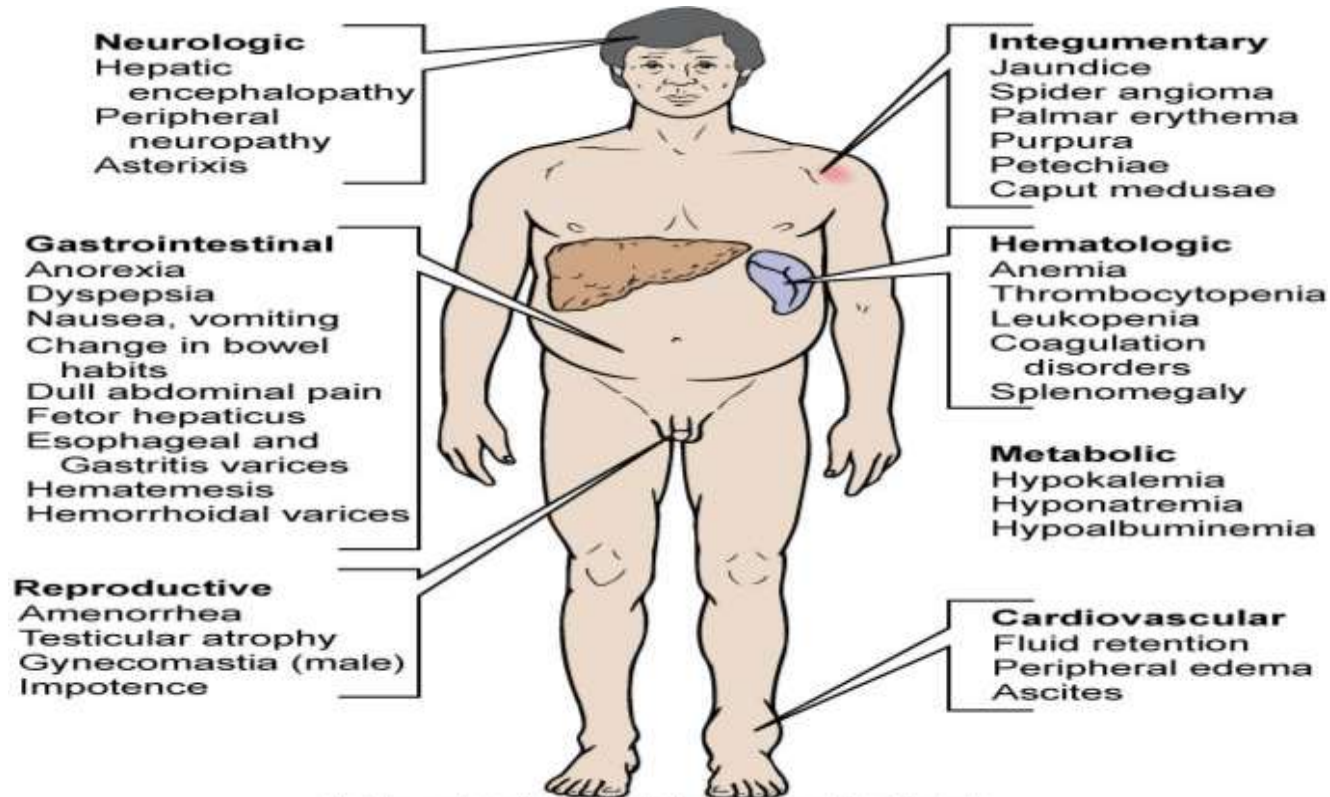


A normal liver shows no signs of scarring



In cirrhosis, scar tissue replaces normal liver tissue.

Clinical Manifestations of liver cirrhosis



GYNECOMASTIA



Dupuytren contracture



PALMAR ERYTHEMA



SPIDER NEAVI



Ascites & clubbing



Complications of liver cirrhosis

Major Complications

- Ascites
- Spontaneous bacterial peritonitis
- Hepatic encephalopathy
- Portal hypertension
- Variceal bleeding
- Renal failure
- Portal vein thrombosis
- Hepatocellular carcinoma
- Hemorrhagic manifestations





Liver Function Tests

Total Bilirubin (0.2 – 1.2 mg/dL) .A breakdown product of hemoglobin (orange-yellow pigmented). bilirubin is processed in the liver. unconjugated and conjugated bilirubin are increased in liver disease

Albumin (3.5 – 5.3 g/dL) . A protein made by the liver. Low levels indicate liver disease

Prothrombin Time (PT) (10 – 14 seconds) .Evaluates the overall ability to produce a clot in a reasonable amount of time.

Platelet Count (150,000 – 450,000 mm³) .Thrombocytopenia is a decrease in the number of circulating platelets. Platelets are not a part of the coagulation cascade, but are essential for the initiation of hemostasis





Other investigations

AST (aspartate aminotransferase) (11 – 47 IU/L).

ALT (alanine aminotransferase) (7 – 56 IU/L) ALT is a useful test for detecting liver damage. •

The AST/ALT ratio is usually increased in alcoholic hepatitis

Alkaline phosphatase (ALP) (30 – 120 IU/L) Found in bone and in the cells of bile ducts. ALP can indicate blockage of one or more bile ducts, liver cancer, hepatitis, cirrhosis, or when hepatotoxic drugs are taken

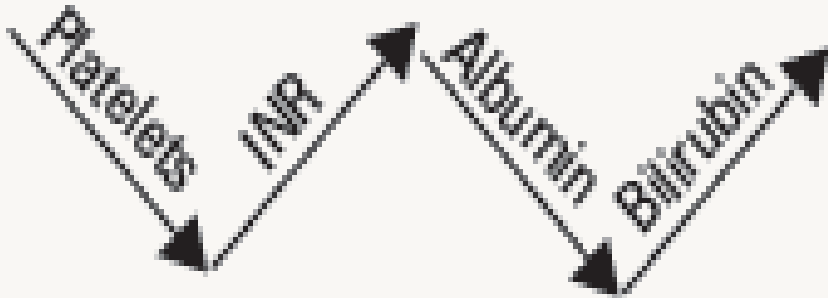




Liver Function Tests

Progression of liver dysfunction based on liver function tests –

The “W”





Other investigations

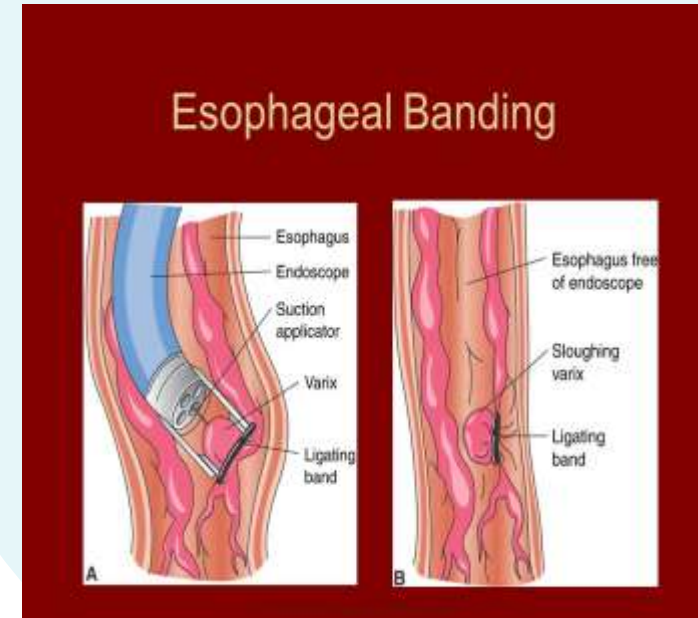
**Definitive diagnosis is histological
(liver biopsy)**

- Serum α -fetoprotein if > 400 ng/mL
hepatocellular carcinoma.



Other investigations

- **Ultrasound :**
Size, shape and echogenicity of the liver.
- **CT scan .**
hepatocellular carcinoma.
- **Endoscopy: diagnostic and therapeutic for** varices, and portal hypertensive gastropathy.
- **MRI scan.**
- **Elastography**



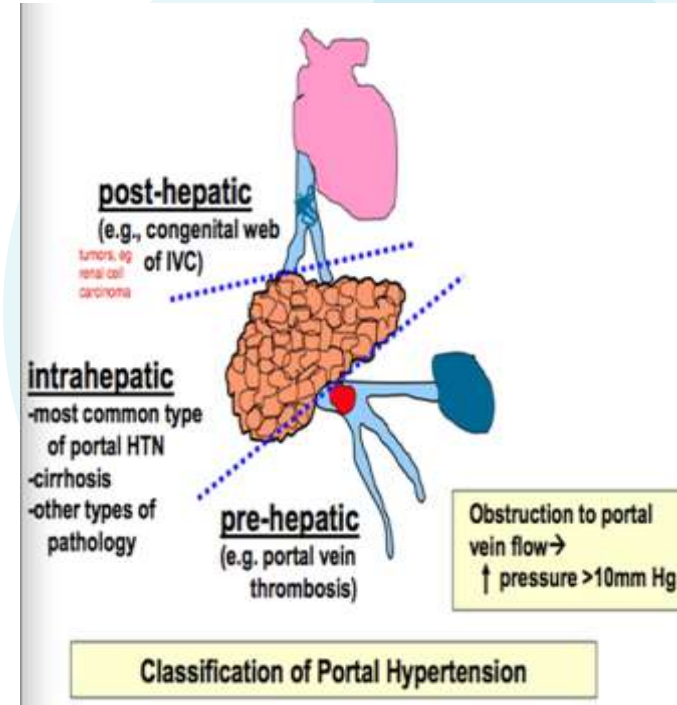
Portal hypertension

Portal hypertension can be classified according to the site of obstruction:

❑ **Pre-hepatic** - due to blockage of the portal vein before the liver as in Portal vein thrombosis

❑ **Intra-hepatic** - due to distortion of the liver architecture as in cirrhosis and schistosomiasis

❑ **Pos-hepatic** - due to venous blockage outside the liver (rare) as in Budd-Chiari syndrome, Veno-occlusive disease, Right heart failure and in Constrictive pericarditis





Clinical features

- Patients with portal hypertension are often asymptomatic and the **only clinical evidence of portal hypertension is splenomegaly.**
- Clinical features of chronic liver disease are usually present .
- Presenting features may include:
 - haematemesis or melaena from rupture of gastro-oesophageal varices or portal hypertensive gastropathy
 - ascites
 - Encephalopathy

Oral Manifestations of liver cirrhosis

Poor hygiene and (caries) are prominent oral findings in patients with chronic alcoholism.

Patients with cirrhosis have impaired gustatory function and are **malnourished**.

Nutritional deficiencies can result in **glossitis and loss of tongue papillae, angular or labial cheilitis**, which is complicated by concomitant candidal infection

Vitamin **K deficiency**, disordered hemostasis, portal hypertension, and **splenomegaly** (causing **thrombocytopenia**) can produce spontaneous gingival bleeding, mucosal ecchymoses, and petechiae.

A sweet, musty odor to the breath is associated with liver failure and jaundiced mucosal tissue
Bilateral, painless hypertrophy of the parotid glands (sialadenosis) is a frequent finding in
Patients with cirrhosis

plaque, calculus, and gingival inflammation more common than those without the condition.



Dental Considerations In liver cirrhosis

- CBC with differential, AST and ALT, bleeding time, thrombin time, and prothrombin time are sufficient in **screening** for potential problems. and should be ordered before invasive procedures
- Abnormal laboratory values, abnormal clinical examination or a positive history, provide the basis for **referral** to a physician
- A patient with **untreated alcoholic** liver disease is **not a candidate for elective**, outpatient dental care and should be referred to a physician.
- Once the patient has been managed medically, dental care may be provided after consultation with the physician



Dental Considerations In liver cirrhosis

- PT/INR L platelet count requirements for surgery is
- **Maximum INR 3.5 and minimum platelet 50000**
- Increased prothrombin time may indicate use of
- local haemostatic agents
- fresh frozen plasma
- vitamin K
- platelets
- antifibrinolytic agents.
- Haemostatic measures are particularly important when major invasive or traumatic procedures may need



Dental Considerations In liver cirrhosis

- Local Anesthesia in normal doses are safe but the maximum dose is lower than in healthy
- **Articaine** is preferred to lidocaine
- Consult a **specialist** for drug doses and preoperative preparation if need surgery
- Avoid **hepatotoxic** drugs
- Avoid **anticoagulant** that can cause bleeding
- Reduced doses of **acetaminophen** or **codeine** for analgesia
- Avoid **NSAID** as increase risk of GIT bleeding
- **Penicillin, amoxicillin, erythromycin, cephalosporins, imipenem** can be used



Dental Considerations In liver cirrhosis

- Decrease dose of midazolam (sedation)
- Inhalation sedation is preferred over iv sedation with benzodiazepine
- Nitrous oxide and pethidine can be used
- Avoid halothane as it can cause hepatitis
- Surgery in jaundice patient: IV fluids, diuretics and mannitol to prevent Acute renal failure (hepatorenal syndrome)
- Prophylaxis against spontaneous bacterial peritonitis:
 - **amoxacillin 2 g + metronidazole 1 hour before surgery**
or IV imipenem



Dental Management of Liver Transplant

Pre-transplant

- Comprehensive dental evaluation
- Extraction of infected, non-restorable, or periodontally hopeless teeth.
- Oral hygiene instruction

Post-transplant

No elective dental treatment for 3 months following surgery

- Routine antibiotics prophylaxis is not recommended Recall program after 3 months



Dental Management of Liver Patients

Dental Drugs

Analgesics/Pain Control

- Aspirin, ibuprofen, and other NSAIDs – **use caution**
- Acetaminophen – **use caution**
- Narcotics – **increase dose interval, short term**
- Morphine – **safe**

Anesthetics

- Lidocaine, mepivacaine – **limit to 300 mg max dosage**
- Prilocaine – **limit to 400 mg max dosage**
- Articaine – **safe (metabolized in plasma)**

Sedatives/Anxiolytics

- Benzodiazepines – **reduce dosage, increase intervals**

Antibiotics

- Beta-lactam (penicillins, ampicillin, cephalexin, cefazolin, ceftriaxone) – **safe (renal excretion)**
- Metronidazole – **interaction w/ alcohol**
- Clindamycin, aminoglycosides – **use caution**
- Tetracyclines – **reduced dosage, increase intervals**

BOX 10-3 Dental Drugs Metabolized Primarily by the Liver

Local Anesthetics*

Lidocaine (Xylocaine)
Mepivacaine (Carbocaine)
Prilocaine (Citanest)
Bupivacaine (Marcaine)

Analgesics

Aspirin[†]
Acetaminophen (Tylenol, Datril)[†]
Codeine[†]
Meperidine (Demerol)[†]
Ibuprofen (Motrin)[†]

Sedatives

Diazepam (Valium)[†]
Barbiturates[†]

Antibiotics

Ampicillin
Tetracycline
Metronidazole[†]
Vancomycin[†]

Type of drug	Drugs contraindicated	Alternatives to use
Analgesics	Aspirin Codeine Dextropropoxyphene Indometacin Mefenamic acid Meperidine ^a NSAIDs Opioids Pentazocine	Acetaminophen/ paracetamol ^a Oxycodone
Antimicrobials	Aminoglycosides Azithromycin Azole antifungals (miconazole, ketoconazole, itraconazole) Clarithromycin ^a Clindamycin ^a Co-amoxiclav* Co-trimoxazole Doxycycline Erythromycin estolate Flucloxacillin Metronidazole ^a Minocycline Moxifloxacin Roxithromycin Salampicillin Tetracyclines	Amoxicillin Ampicillin, cephalosporins Nystatin, fluconazole Erythromycin stearate Imipenem Penicillin Nystatin Penicillin Tetracycline Penicillin

And liver



Type of drug	Drugs contraindicated	Alternatives to use
Corticosteroids	Prednisone	Prednisolone
Antidepressants	Monoamine oxidase inhibitors Tricyclics	SSRIs
Muscle relaxants	Suxamethonium	Atracurium, cisatracurium, pancuronium, vecuronium
Local anaesthetics	Lidocaine ^a	Articaine, prilocaine
General anaesthetics	Halothane Methohexitone Propofol ^a Thiopental	Desflurane Isoflurane Sevoflurane
Anxiolytics/ sedatives	Barbiturates Diazepam ^a Midazolam ^a (and flumazenil) Phenothiazines Promethazine	Lorazepam ^a Oxazepam ^a Pethidine ^a



And liver

Type of drug	Drugs contraindicated	Alternatives to use
Anticonvulsants	Carbamazepine Lamotrigine Phenytoin	
Others	Anticoagulants Biguanides Diuretics Etreinate Liquid paraffin Lomotil Methyldopa Oral contraceptives Pilocarpine Sumatriptan	

And liver



Patient Evaluation/Risk Assessment (see Box 1-1)

- Evaluation is directed at determining the nature, severity, control, and stability of disease.

Potential Issues/Factors of Concern

A	
Analgesics	Nonsteroidal antiinflammatory drugs (NSAIDs), including aspirin, and acetaminophen, as well as codeine and meperidine, should be avoided or their use very limited in persons who have end-stage liver disease.
Antibiotics	Antibiotic prophylaxis is not recommended; however, patients who have severe liver disease may be more susceptible to infection. Selection of antibiotic agent is based on risk and severity of dental infection. Avoid use of metronidazole and vancomycin.
Anesthesia	Higher doses may be required to achieve adequate anesthesia in presence of alcoholic liver disease. Knowledge of current liver function is important to establish proper dosages. Epinephrine (1:100,000, in a dose of no more than two carpules) in local anesthetics generally is not associated with any problems, but patients should be monitored closely.
Anxiety	Use anxiety/stress reduction techniques as needed, but avoid benzodiazepines.
Allergy	No issues.
B	
Breathing	No issues.
Bleeding	Excessive bleeding may occur in the patient with end-stage liver disease. Most such patients will have reductions in coagulation factors and thrombocytopenia, so they are at greater risk for postsurgical bleeding; they may need vitamin K and/or platelet or clotting factor replacement.

Blood pressure

Monitor blood pressure, I

C	
Chair position	No issues.
Consultation	Once the patient is under good medical management, the dental treatment plan is unaffected. However, consultation with the patient's physician to establish the level of control and to identify bleeding tendencies and altered drug metabolism is recommended as part of the management program.
D	
Devices	No issues.
Drugs	Because many medications are metabolized in the liver, certain drugs may need to be avoided or reduced in dosage. Limit or avoid use of acetaminophen, aspirin, ibuprofen, codeine, meperidine, diazepam, barbiturates, metronidazole, and vancomycin. Refer to a good drug reference. The use of epinephrine or other pressor amines (in gingival retraction cord or to control bleeding) must be limited, especially if portal hypertension is present.
E	
Equipment	No issues.
Emergencies and urgent care	For patient with severe liver disease who requires urgent care, consider treating in special care clinic or hospital. After consulting with physician, provide limited care only for pain control, treatment of acute infection, or control of bleeding until condition improves.
F	
Follow-up	It is important to follow up with the patient post-operatively to be certain that there are no complications.

Dental Drugs Metabolized Primarily by the Liver

Local anesthetics (appear safe for use during liver disease when used in appropriate amounts)

- Lidocaine (Xylocaine)
- Mepivacaine (Carbocaine)
- Prilocaine (Citanest)
- Bupivacaine (Marcaine)

Analgesics

- Aspirin *
- Acetaminophen (Tylenol, Datril) †
- Codeine †
- Meperidine (Demerol) †
- Ibuprofen (Motrin) *

Sedatives

- Diazepam (Vallium) †
- Barbiturates †

Antibiotics

- Ampicillin
- Tetracycline
- Metronidazole ‡
- Vancomycin ‡

• (*) Limit dose or avoid if severe liver disease (acute hepatitis and cirrhosis) or hemostatic abnormalities are present.

• (†) Limit dose or avoid if severe liver disease (acute hepatitis and cirrhosis) or encephalopathy is present, or if taken with alcohol.

• (‡) Avoid if severe liver disease (acute hepatitis and cirrhosis) is present.



Liver Cirrhosis and Dental Management – Case-Based Discussions

A 52-year-old male with Hepatitis C-induced cirrhosis

- Dental Procedure:** Extraction of tooth

- Medical Info:**

INR: 2.0, Platelets: 65,000/mm³

On lactulose, propranolol, mild ascites, no encephalopathy

1. What are the risks of performing extraction?
2. How does cirrhosis affect coagulation?
3. What lab values should be assessed pre-op?
4. Would you consult the physician first?



Liver Cirrhosis and Dental Management – Case-Based Discussions

- INR >1.5 and thrombocytopenia increase bleeding risk.
- Consult hepatologist – may need platelet transfusion or FFP.
- Use local hemostatics: sutures, tranexamic acid, collagen sponge.
- Avoid NSAIDs post-op. Monitor healing.



Liver Cirrhosis and Dental Management – Case-Based Discussions

45-year-old female with alcoholic liver cirrhosis. Prescribed antibiotics and analgesics after root canal treatment

Liver Function Tests showed Elevated bilirubin, INR 1.5, albumin low

You give him Amoxicillin + Paracetamol

Discussion Points:

1. How does liver dysfunction affect drug metabolism?
2. Is paracetamol safe in liver disease? What is the safe dose?
3. Should NSAIDs be avoided? Why?
4. Would you change the antibiotic choice?



Liver Cirrhosis and Dental Management – Case-Based Discussions

Paracetamol: Safe at $\leq 2\text{g/day}$ short-term

Avoid use of NSAD due to bleeding and renal risk

Clindamycin or metronidazole OK with caution ,Use lowest effective dose, monitor liver symptoms



Liver Cirrhosis and Dental Management – Case-Based Discussions

Patient: 55-year-old male with decompensated cirrhosis (Child-Pugh class C)

Dental Need: Full-mouth rehabilitation with extractions and prosthetics

Concerns: Ascites, encephalopathy episodes, bleeding risk, poor wound healing

Discussion Points:

1. Should elective surgery be deferred?
2. What are the criteria for medical clearance?
3. Can prosthetic rehabilitation be done with minimal surgical intervention?



Liver Cirrhosis and Dental Management – Case-Based Discussions

Elective invasive procedures are contraindicated in decompensated patients
Increased risk of hemorrhage, infection, poor wound healing
Defer until stabilization or coordinate in hospital setting
Focus on conservative, palliative care temporarily



Liver Cirrhosis and Dental Management – Case-Based Discussions

Patient: 48-year-old male with Hepatitis B

Dental Visit: Routine scaling

Viral Load: Undetectable

Dentist Concern: Cross-infection risk

- 1. What are the precautions should be done?**
- 2. Is it ethical to deny treatment?**
- 3. What if the viral load was high?**



Liver Cirrhosis and Dental Management – Case-Based Discussions

Use standard precautions for all patients

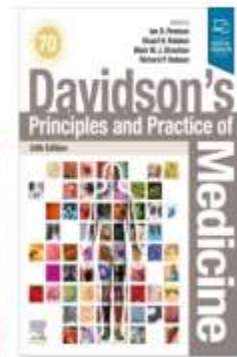
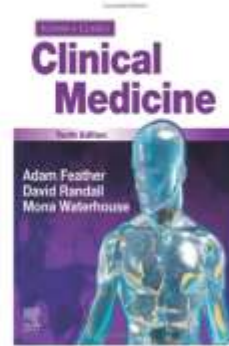
No need for extra isolation with undetectable load

Ethically and legally, must not discriminate

High viral load: consult specialist, consider enhanced precautions if surgical

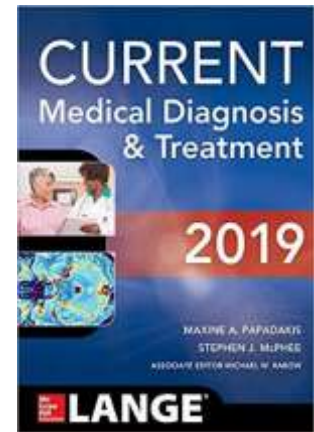
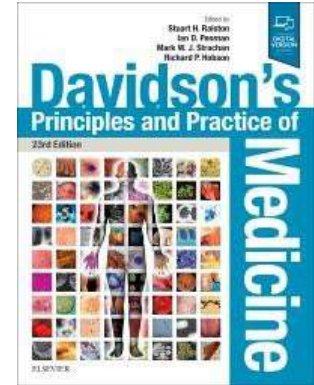
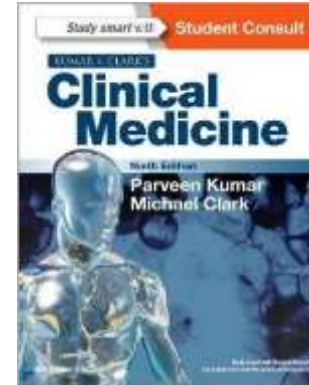
References & Further Readings

- **Kumar and Clark's Clinical Medicine**, 10th Edition
- **Davidson's Principles and Practice of Medicine**, 24th Edition
- **Oxford Handbook of Clinical Medicine**, 10th Edition
- **CURRENT Medical Diagnosis and Treatment 2024 (LANGE CURRENT Series)** 63rd Edition



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THANKS!

